



Delivery Delay Prediction in Logistics Systems

"Model Comparison" Results

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Model Comparison

Champion	Name	Algorithm Name	KS (Youden)	Accuracy
true	Forest	Forest	0.4858	0.6891
false	Logistic Regression	Logistic Regression	0	0.5964
false	Gradient Boosting	Gradient Boosting	0.4834	0.6736

Average Squared Error	Area Under ROC	Cumulative Lift	Cumulative Captured Response Percentage	Cutoff
0.1771	0.7602	1.6768	16.7683	0.5000
0.2407	0.5000	1	10	0.5000
0.1793	0.7545	1.6768	16.7683	0.5000

Data Role	Depth	F1 Score	False Discovery Rate	False Positive Rate
TEST	10	0.7036	0.1847	0.2072
TEST	10	0.7472	0.4036	1
TEST	10	0.6908	0.2059	0.2342

Gain	Gini Coefficient	ROC Separation	Lift	Misclassification Rate
0.6768	0.5203	0.4117	1.6768	0.3109
0.0000	0	0	1	0.4036
0.6768	0.5090	0.3770	1.6768	0.3264

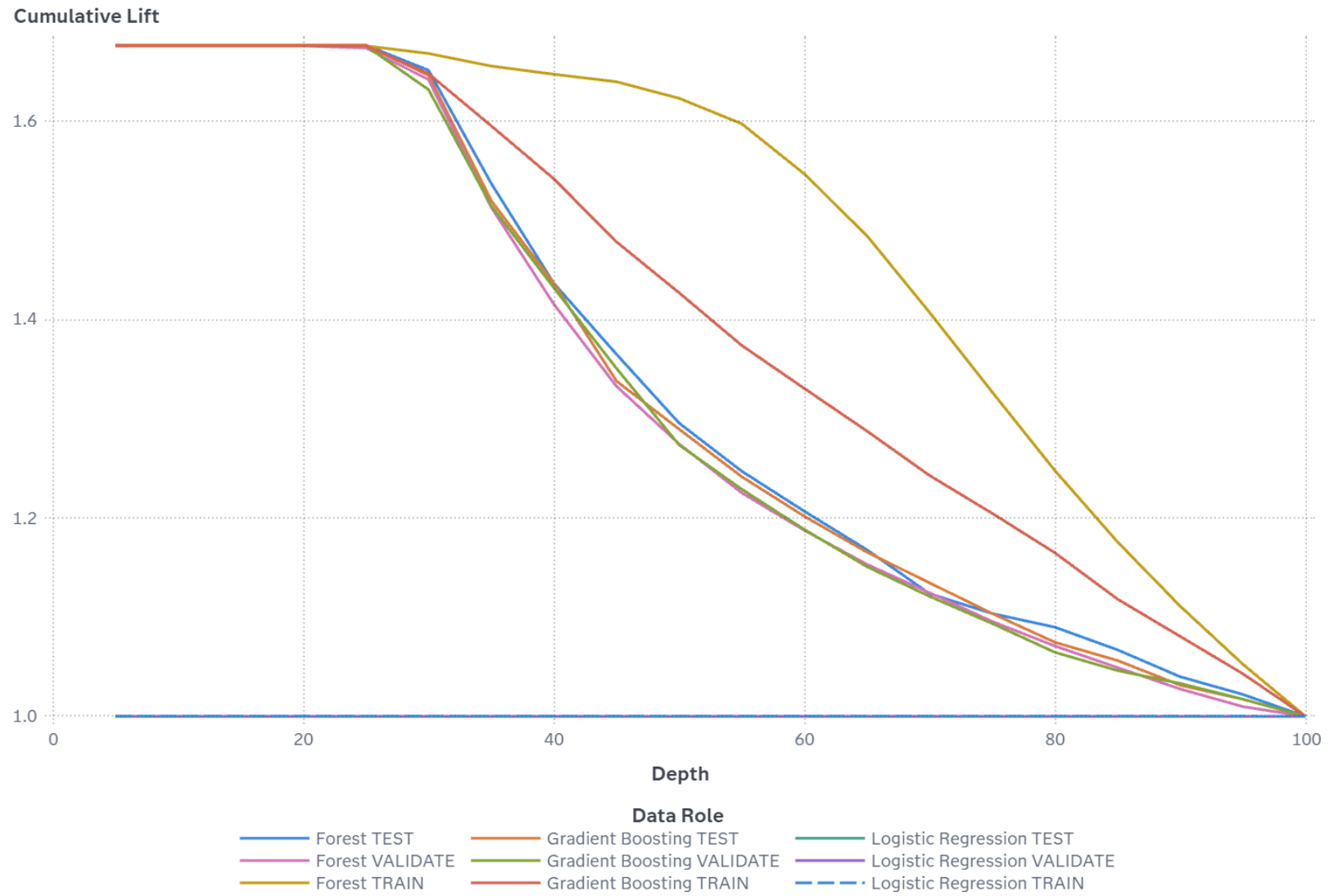
Multi-Class Log Loss	Misclassification at Cutoff	Misclassification Rate (Event)	Number of Observations	Root Average Squared Error
0.4975	0.3109	0.3109	1,100	0.4208
0.6745	0.4036	0.4036	1,100	0.4906
0.5021	0.3264	0.3264	1,100	0.4235

Captured Response Percentage
8.3841
5
8.3841

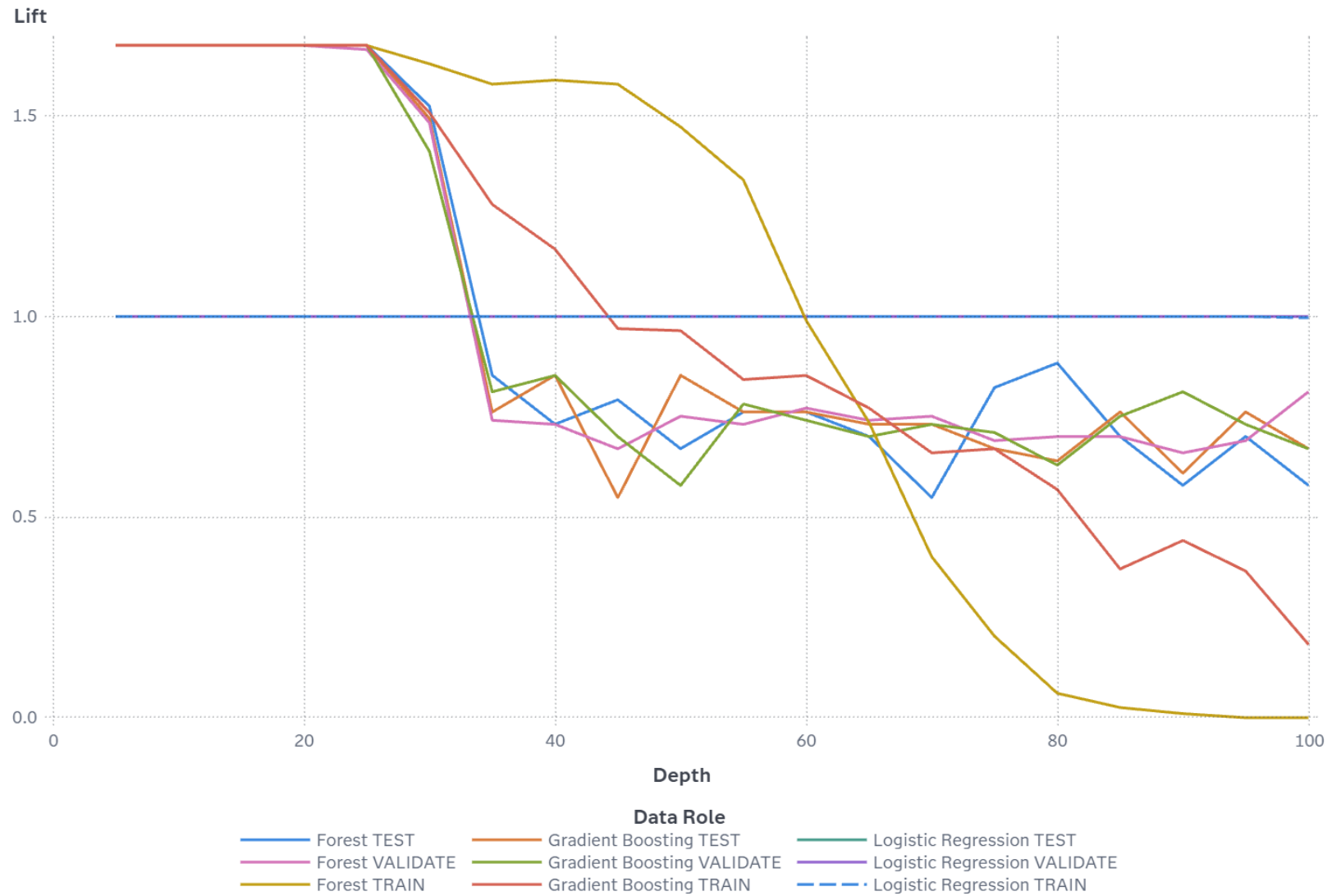
Properties

Property Name	Property Value
selectionCriteriaClass	Kolmogorov-Smirnov statistic (KS)
selectionCriteriaInterval	Average squared error
selectionTable	Test
selectionDepth	10
cutoff	0.50

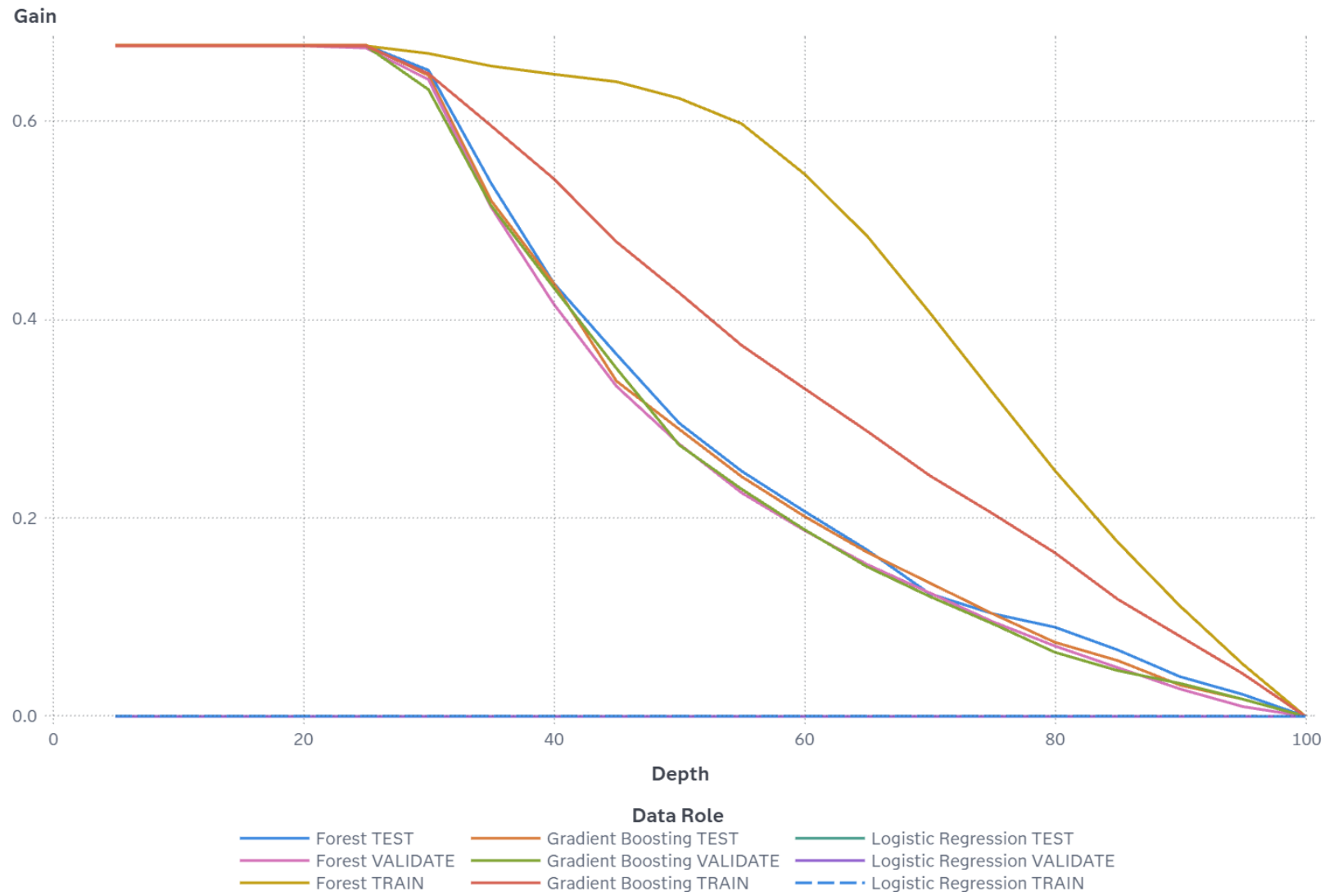
Cumulative Lift



Lift



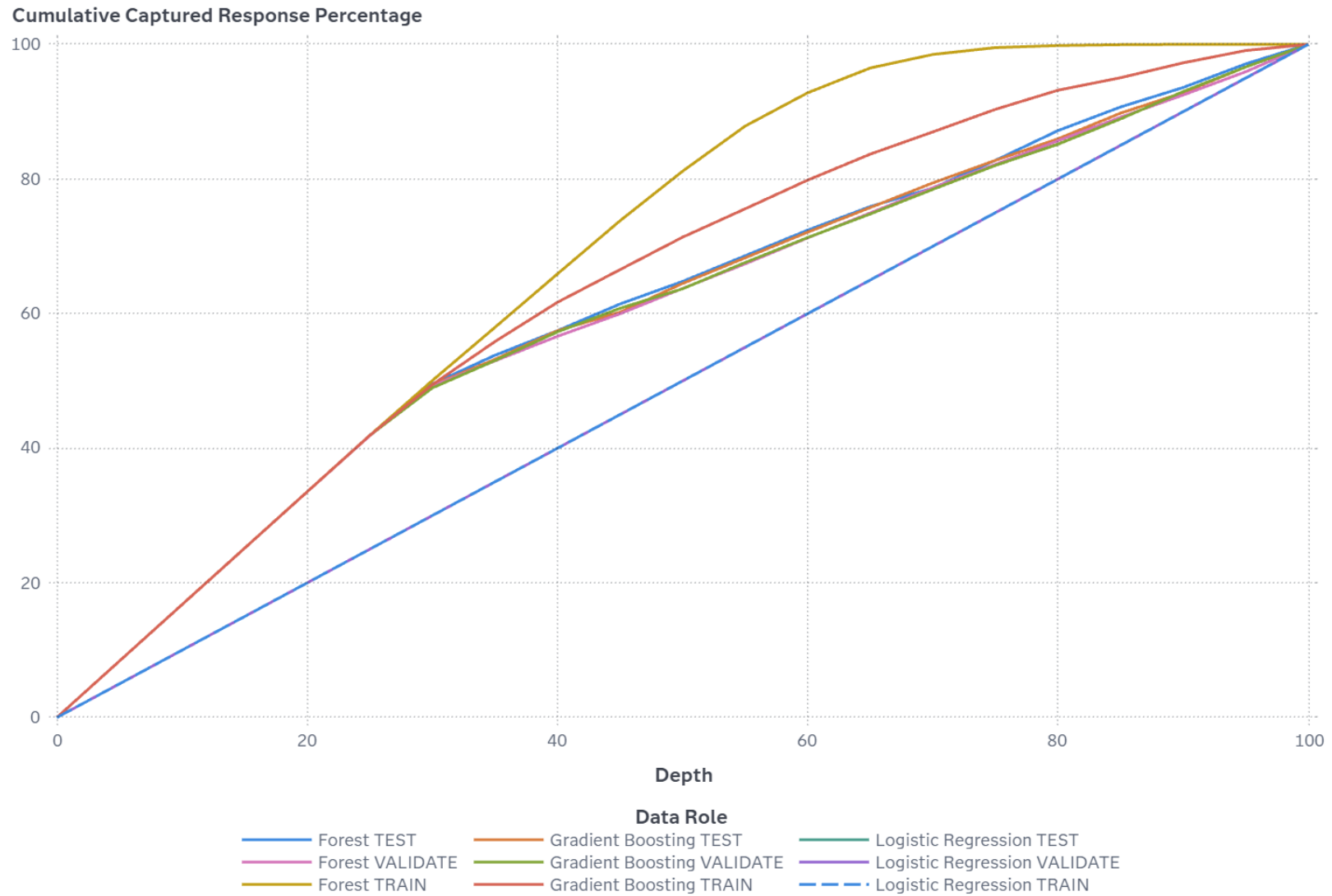
Gain



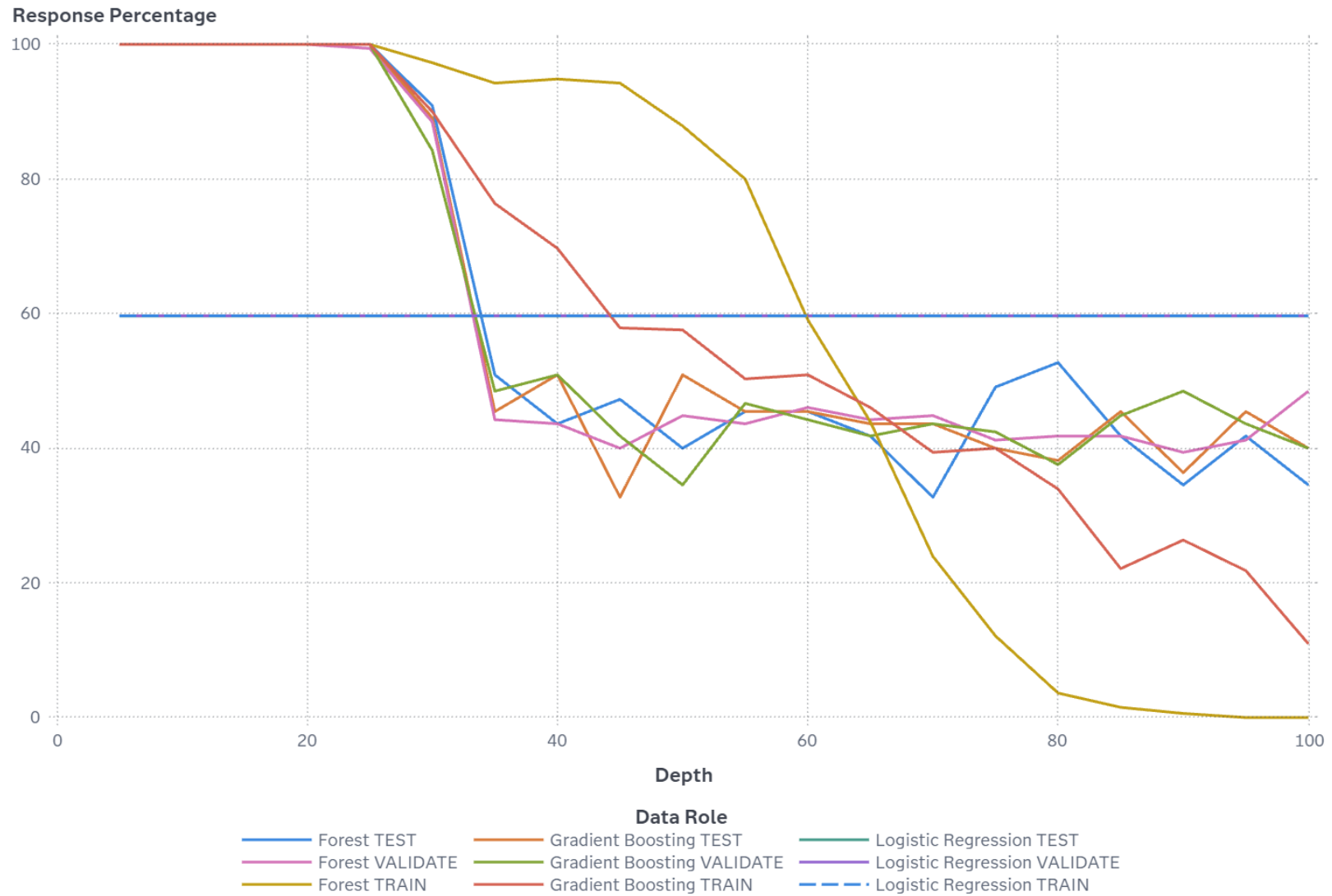
Captured Response Percentage



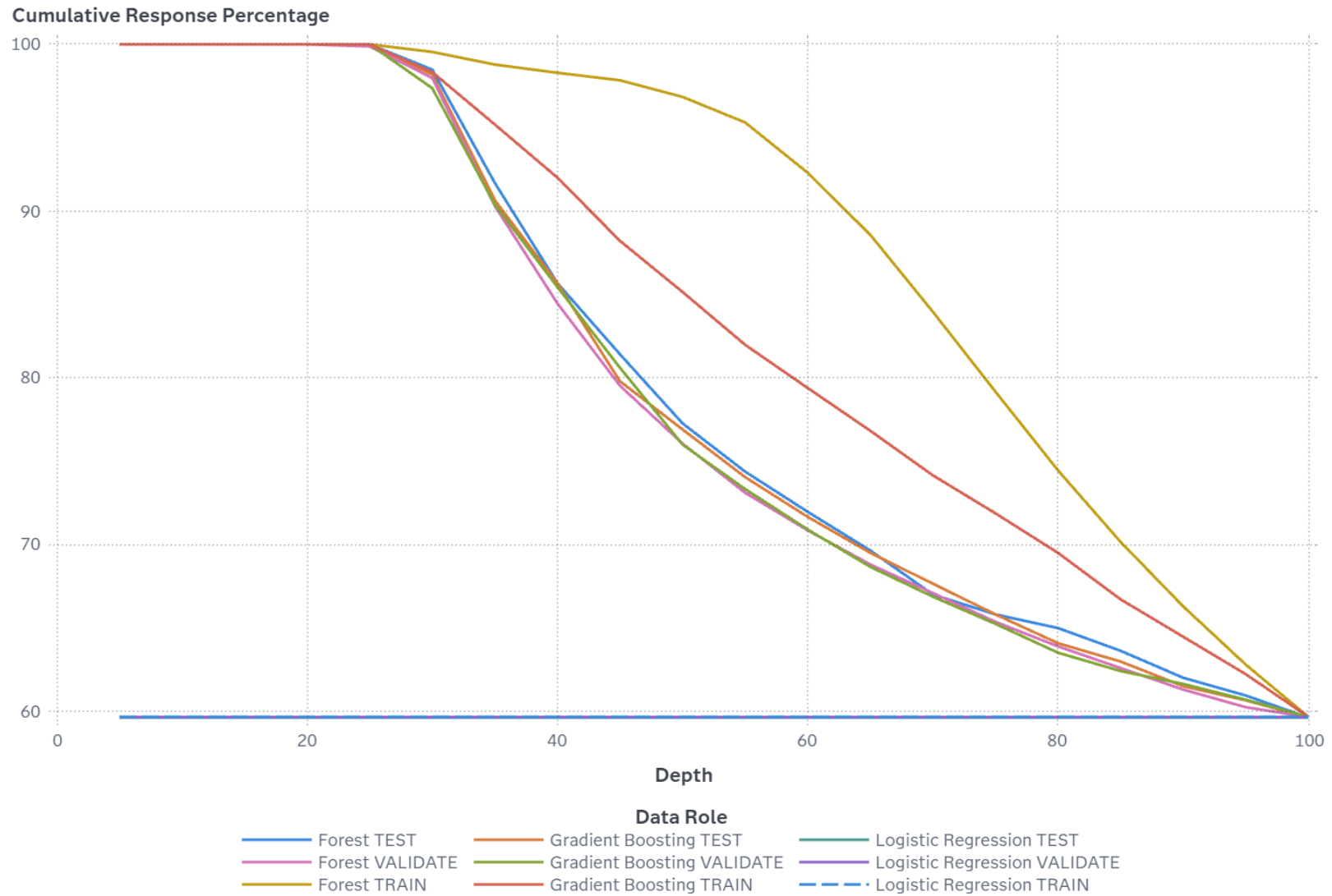
Cumulative Captured Response Percentage



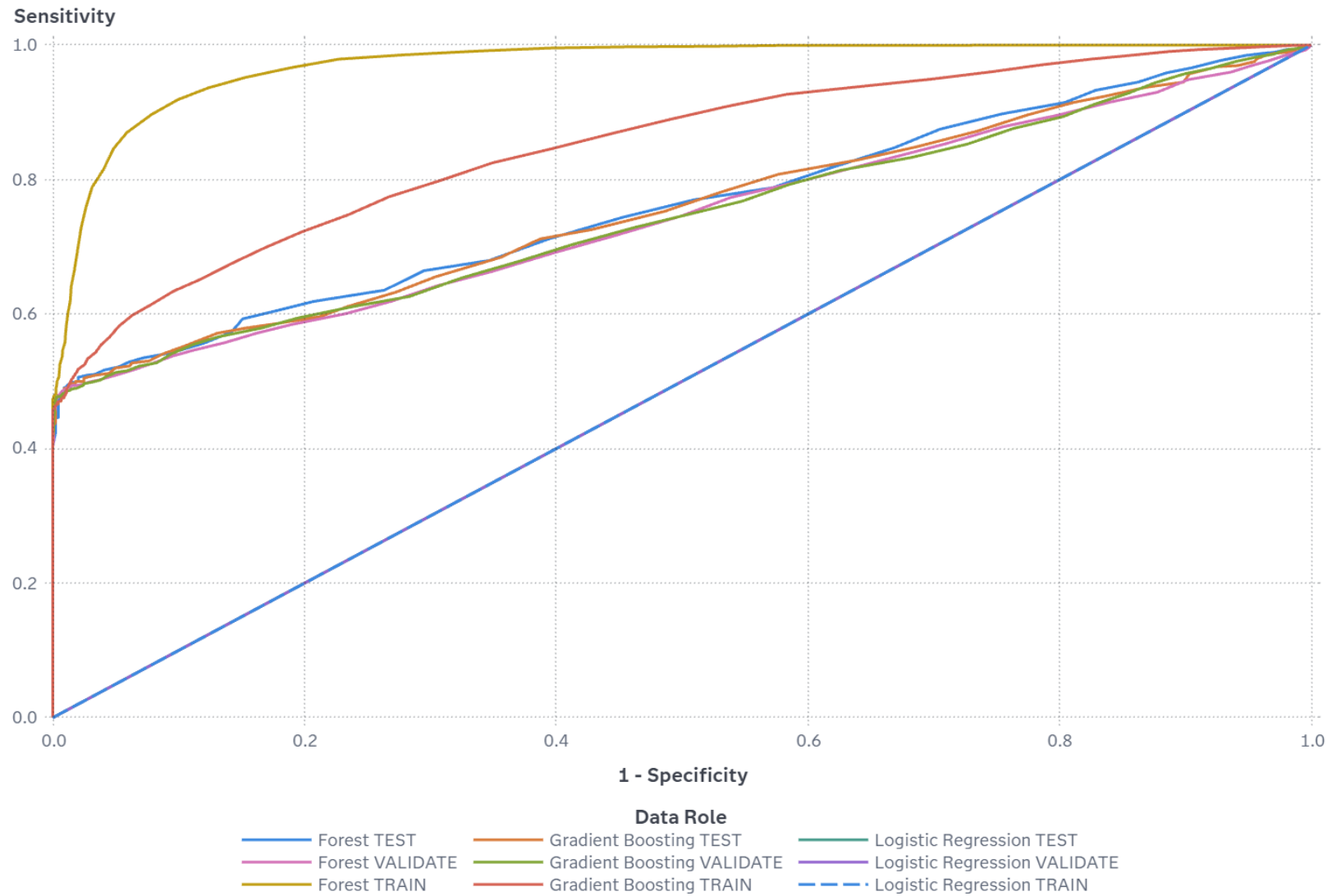
Response Percentage



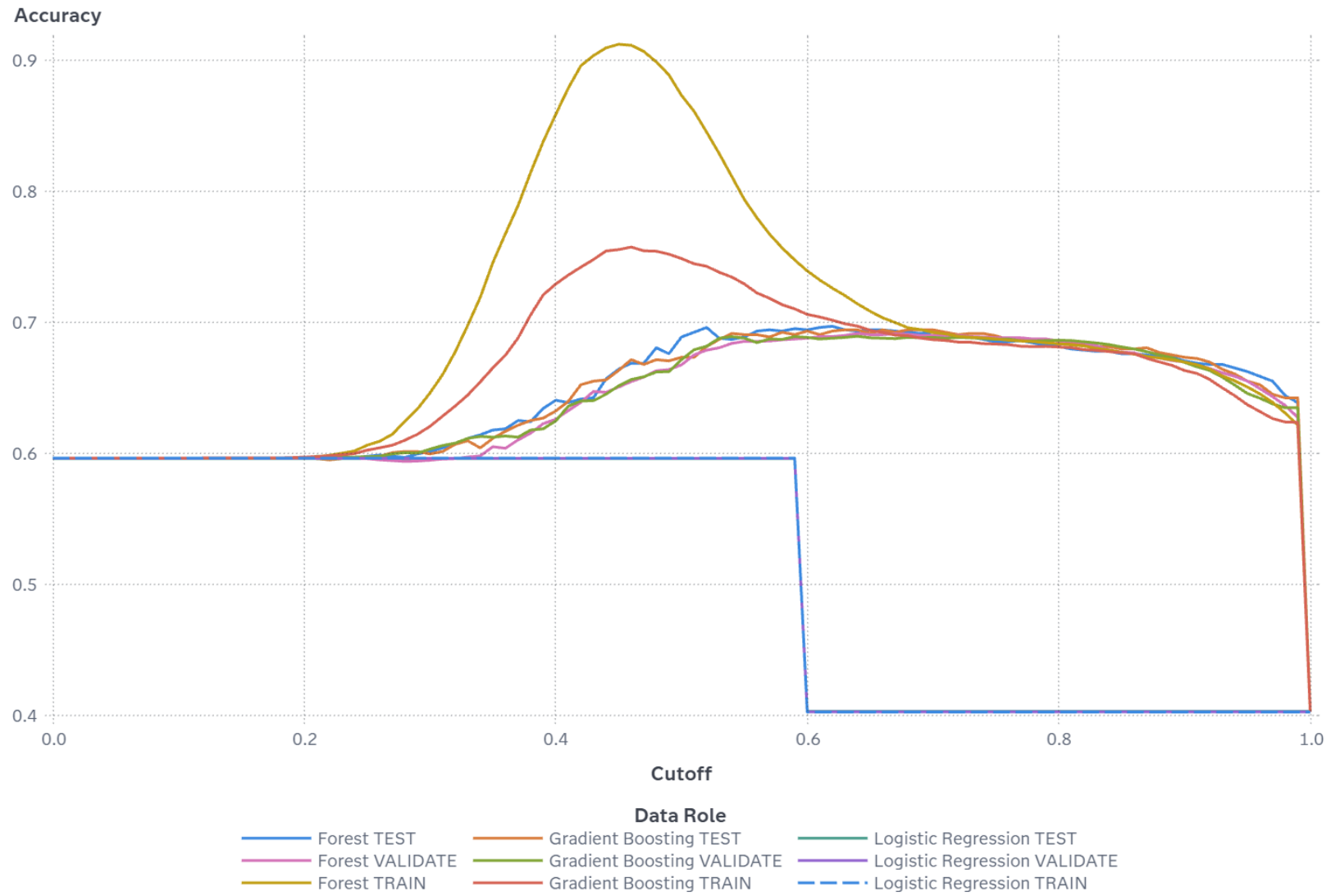
Cumulative Response Percentage



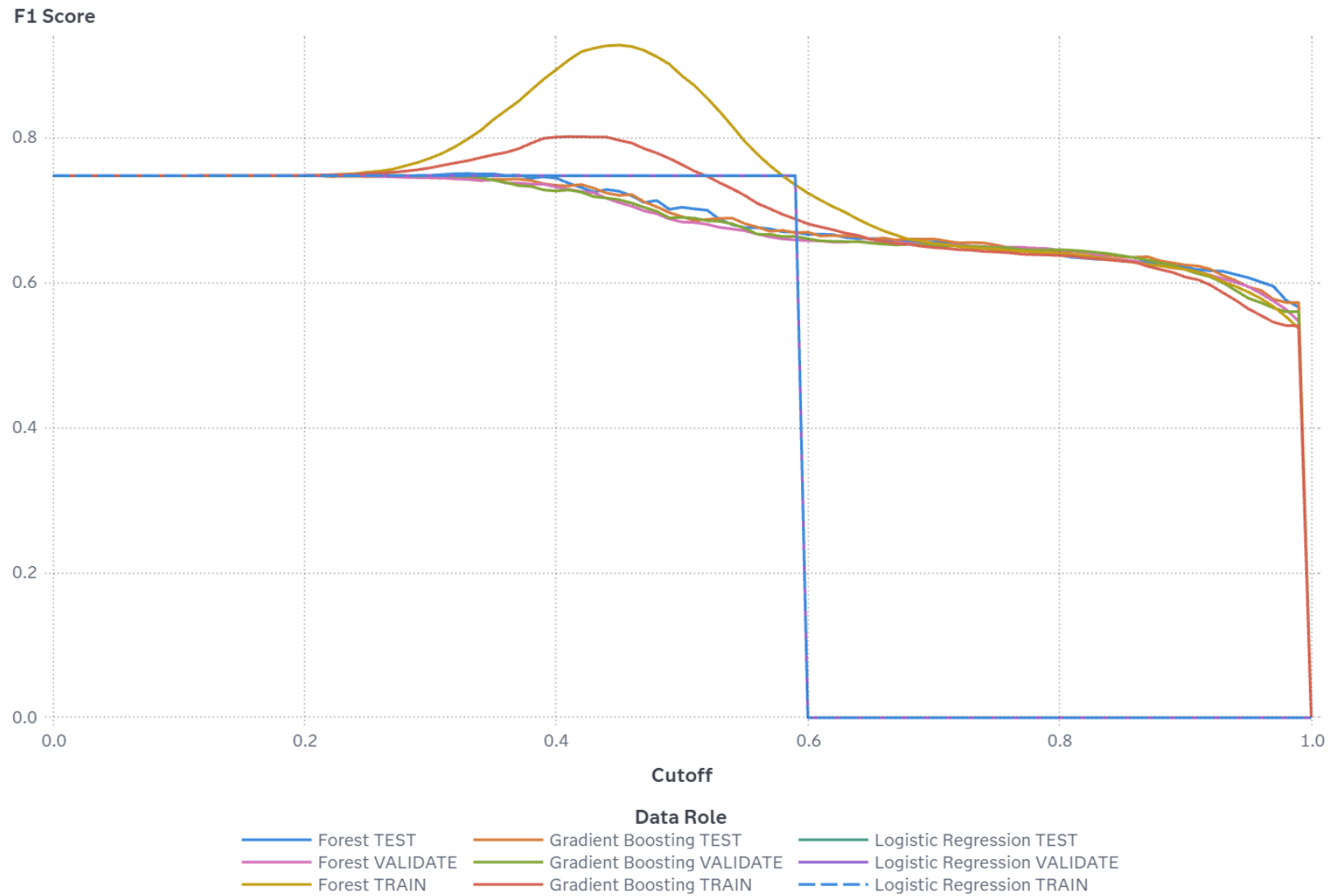
ROC



Accuracy



F1 Score



Fit Statistics

Statistics Label	TRAIN_Forest	VALIDATE_Forest	TEST_Forest	TRAIN_Logistic_Regression
Area Under ROC	0.9715	0.7433	0.7602	0.5000
Average Squared Error	0.1295	0.1804	0.1771	0.2406
Divisor for ASE	6,599	3,300	1,100	6,599
Formatted Partition	1	0	2	1
Gamma	0.9481	0.4960	0.5301	
Gini Coefficient	0.9430	0.4865	0.5203	0
KS (Youden)	0.8195	0.4793	0.4858	0
KS at User-Specified Cutoff	0.7747	0.3679	0.4117	0
KS Cutoff	0.4600	0.6700	0.6200	0
Misclassification Rate	0.1267	0.3321	0.3109	0.4032
Misclassification Rate (Event)	0.1267	0.3321	0.3109	0.4032
Misclassification Rate at KS Cutoff (Event)	0.0885	0.3085	0.3027	0.4032
Multi-Class Log Loss	0.3983	0.5040	0.4975	0.6743
Number of Observations	6,599	3,300	1,100	6,599
Partition Indicator	1	0	2	1

Statistics Label	TRAIN_Forest	VALIDATE_Forest	TEST_Forest	TRAIN_Logistic_Regression
Root Average Squared Error	0.3599	0.4248	0.4208	0.4905
Target Name	Reached.on.Time_Y.N	Reached.on.Time_Y.N	Reached.on.Time_Y.N	Reached.on.Time_Y.N
Tau	0.4539	0.2342	0.2507	0

VALIDATE_Logistic_Regression	TEST_Logistic_Regression	TRAIN_Gradient_Boosting	VALIDATE_Gradient_Boosting	TEST_Gradient_Boosting
0.5000	0.5000	0.8497	0.7452	0.7545
0.2407	0.2407	0.1602	0.1806	0.1793
3,300	1,100	6,599	3,300	1,100
0	2	1	0	2
		0.7102	0.4989	0.5178
0	0	0.6994	0.4905	0.5090
0	0	0.5382	0.4750	0.4834
0	0	0.5333	0.3726	0.3770
0	0	0.5200	0.7200	0.6800
0.4033	0.4036	0.2511	0.3282	0.3264
0.4033	0.4036	0.2511	0.3282	0.3264
0.4033	0.4036	0.2570	0.3112	0.3055
0.6743	0.6745	0.4632	0.5048	0.5021

VALIDATE_Logistic_Regression	TEST_Logistic_Regression	TRAIN_Gradient_Boosting	VALIDATE_Gradient_Boosting	TEST_Gradient_Boosting
3,300	1,100	6,599	3,300	1,100
0	2	1	0	2
0.4906	0.4906	0.4002	0.4250	0.4235
Reached.on.Time_Y.N	Reached.on.Time_Y.N	Reached.on.Time_Y.N	Reached.on.Time_Y.N	Reached.on.Time_Y.N
0	0	0.3367	0.2361	0.2452